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roll no - 25

ML LAB 1

```
import warnings
warnings.filterwarnings('ignore')
```

```
import pandas as pd
df = pd.read_csv('Titanic-Dataset.csv')
display(df.head(20))
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	NaN	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	NaN	C
10	11	1	3	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	PP 9549	16.7000	G6	S
11	12	1	1	Bonnell, Miss. Elizabeth	female	58.0	0	0	113783	26.5500	C103	S
12	13	0	3	Saundercock, Mr. William Henry	male	20.0	0	0	A/5. 2151	8.0500	NaN	S
13	14	0	3	Andersson, Mr. Anders Johan	male	39.0	1	5	347082	31.2750	NaN	S
14	15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14.0	0	0	350406	7.8542	NaN	S
15	16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55.0	0	0	248706	16.0000	NaN	S
16	17	0	3	Rice, Master. Eugene	male	2.0	4	1	382652	29.1250	NaN	Q
17	18	1	2	Williams, Mr. Charles Eugene	male	NaN	0	0	244373	13.0000	NaN	S
				Vander Planke, Mrs.								

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.model_selection import train_test_split
```

```
df.tail()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

```
df.shape
```

```
(891, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
df["Age"].fillna(df["Age"].median(), inplace=True)
```

```
df["Embarked"].fillna(df["Embarked"].mode()[0], inplace=True)
```

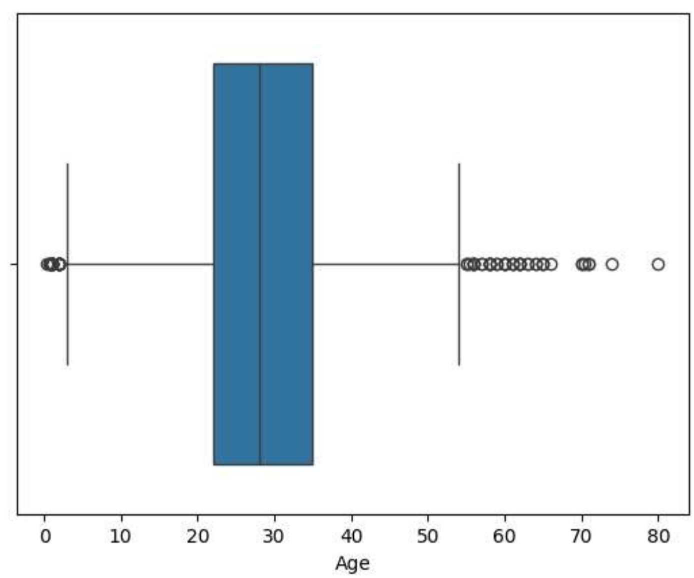
```
df.drop("Cabin",axis = 1 ,inplace = True)
```

```
df.duplicated().sum()
```

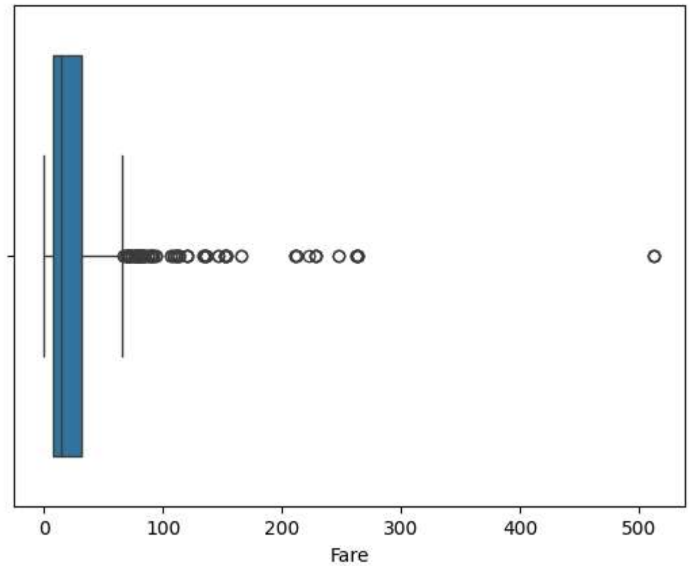
```
np.int64(0)
```

```
df.drop_duplicates(inplace =True)
```

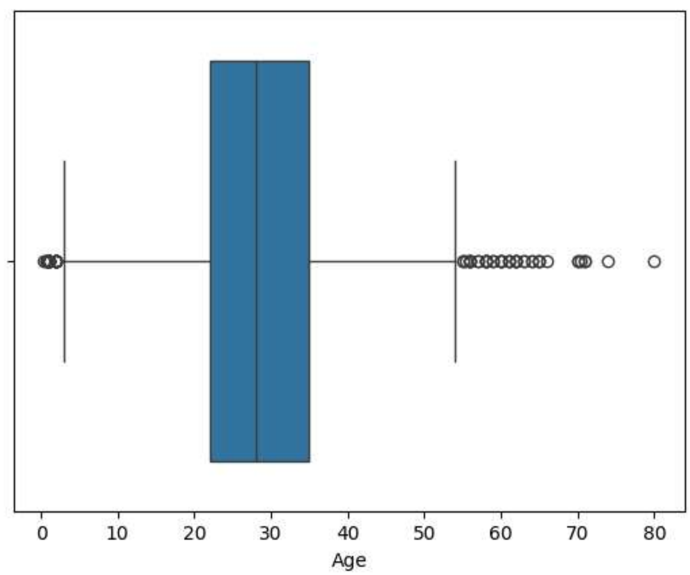
```
sns.boxplot(x=df["Age"])
plt.show()
```



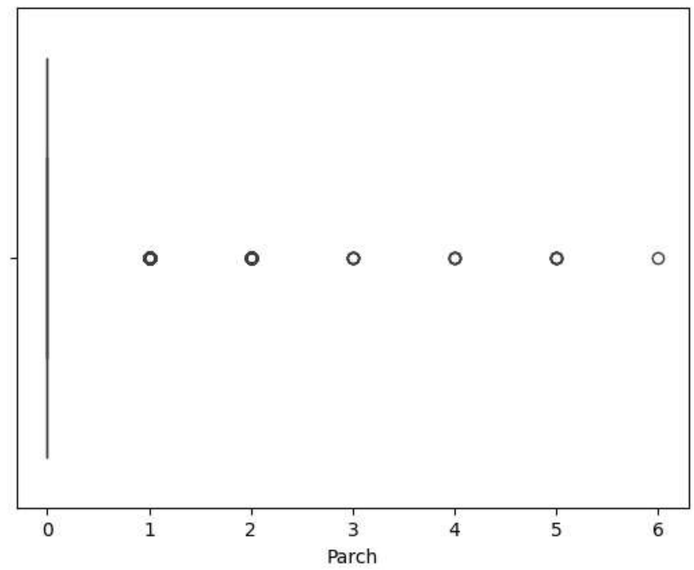

```
sns.boxplot(x=df["Fare"])
plt.show()
```




```
sns.boxplot(x=df["Age"])
plt.show()
```

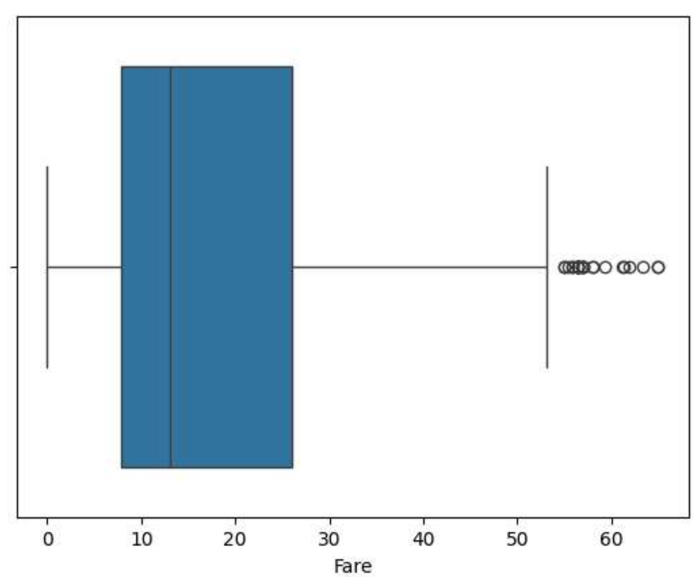



```
sns.boxplot(x=df["Parch"])
plt.show()
```




```
Q1 = df["Fare"].quantile(0.25)
Q3= df["Fare"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

df = df[(df["Fare"] >= lower_bound) & (df["Fare"] <= upper_bound )]
```

```
sns.boxplot(x=df["Fare"])
plt.show()
```



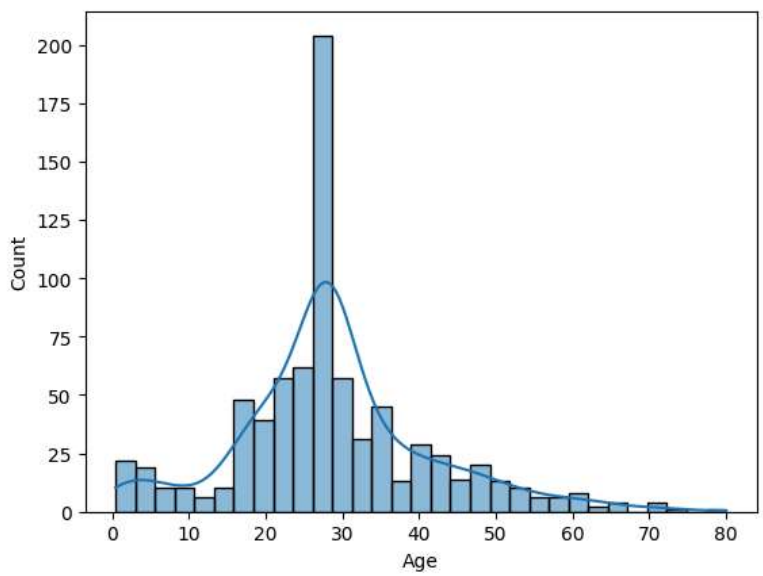

```
le = LabelEncoder()

df["Sex"] = le.fit_transform(df["Sex"])
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500	S
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500	S
5	6	0	3	Moran, Mr. James	1	28.0	0	0	330877	8.4583	Q

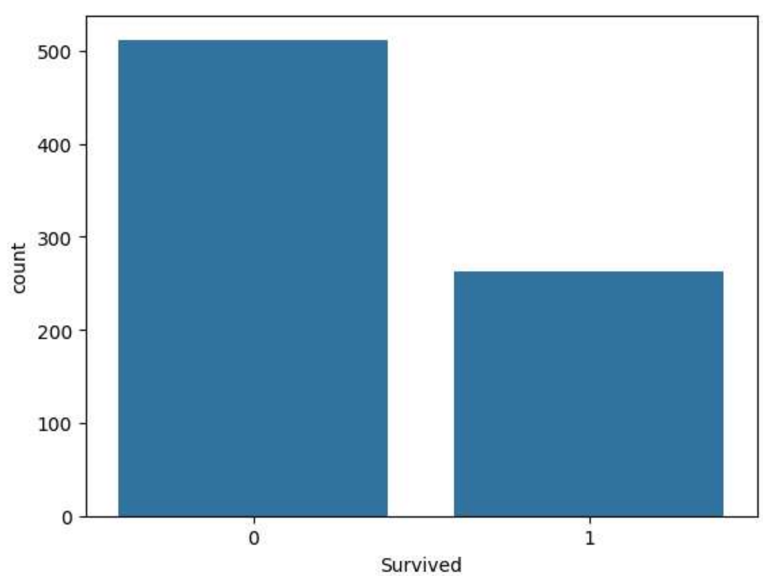
Next steps: Generate code with df New interactive sheet

```
sns.histplot(df["Age"],kde =True)
```

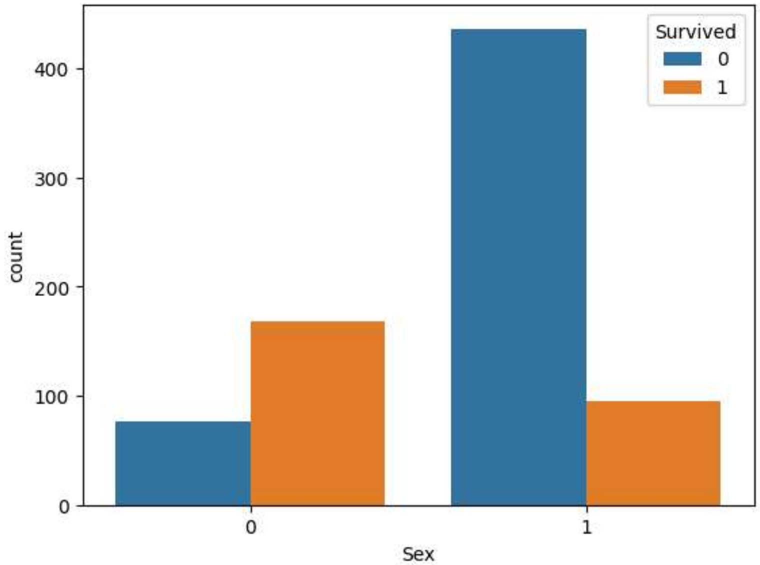
```
<Axes: xlabel='Age', ylabel='Count'>
```



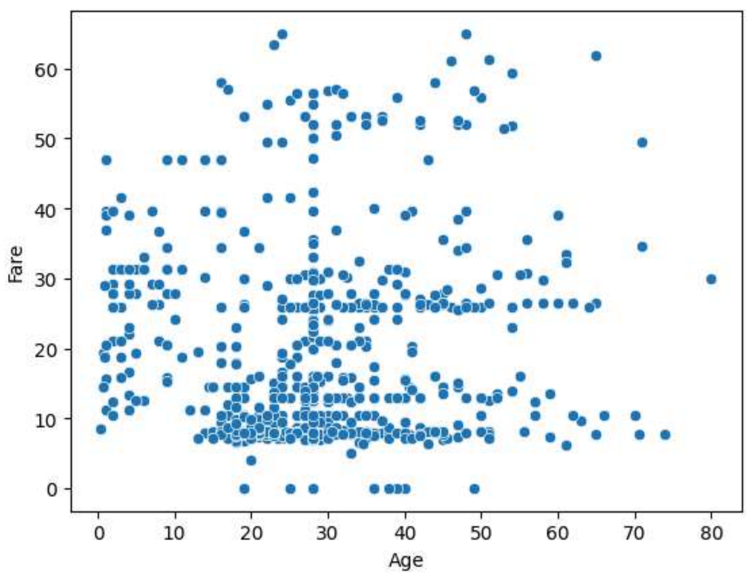

```
sns.countplot(x ="Survived",data=df)

plt.show()
```



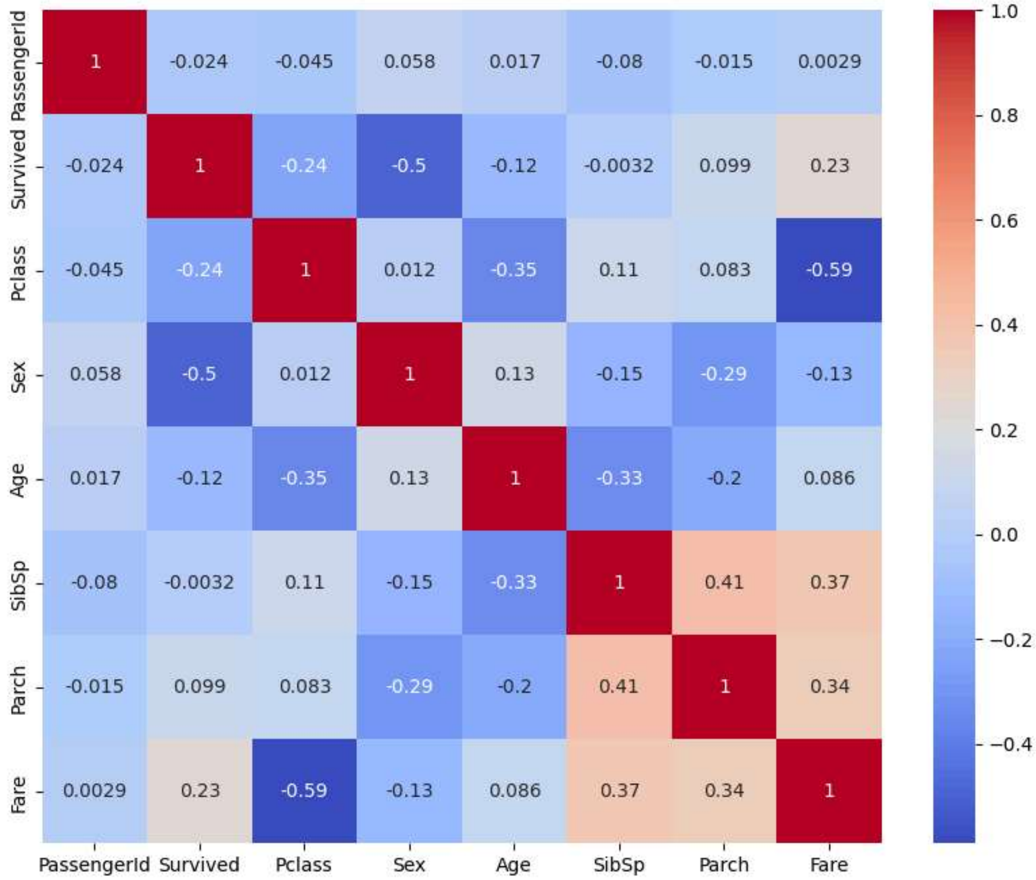

```
sns.countplot(x = "Sex" , hue = "Survived",data = df)
plt.show()
```




```
sns.scatterplot( x = "Age" , y = "Fare" , data = df)
plt.show()
```




```
plt.figure(figsize=(10,8))
numeric_df= df.select_dtypes(include=['number'])
sns.heatmap(numeric_df.corr(),annot=True, cmap='coolwarm')
plt.show()
```




```
scaler = StandardScaler()
df[["Age","Fare"]] = scaler.fit_transform(df[["Age","Fare"]])
```

```
X = df.drop("Survived",axis =1)
y = df["Survived"]
```

```
X = X.drop(["PassengerId","Name","Ticket"],axis =1)
```

```
X_train ,X_test,y_train,y_test = train_test_split(X,y,test_size =0.20,random_state =42)
```

```
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(620, 7)
(155, 7)
(620,)
(155,)
```

```
scaler = StandardScaler()

# Notice the double square brackets instead of .([ ... ])
df[["Age", "Fare"]] = scaler.fit_transform(df[["Age", "Fare"]])
```

Double-click (or enter) to edit

```
x=df.drop("Survived",axis=1)
y=df["Survived"]
```

```
x = x.drop(["PassengerId", "Name", "Ticket"], axis=1, errors='ignore') # useless feature remove
```

```
# sending data for training and testing

X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
```

```
print("Shape of X_train:", X_train.shape)
print("Shape of X_test:", X_test.shape)
print("Shape of y_train:", y_train.shape)
print("Shape of y_test:", y_test.shape)
```

```
Shape of X_train: (620, 7)
Shape of X_test: (155, 7)
Shape of y_train: (620,)
Shape of y_test: (155,)
```

```
x.head(15)
```

	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	3	1	-0.528321	1	0	-0.779117	S
2	3	0	-0.215182	0	0	-0.729373	S
3	1	0	0.489381	1	0	2.599828	S
4	3	1	0.489381	0	0	-0.720161	S
5	3	1	-0.058613	0	0	-0.690071	Q
6	1	1	1.976792	0	0	2.508630	S
7	3	1	-2.094017	3	1	0.239725	S